GE Energy

GEK-SA-1052

ControlST* Software Upgrade for Darwin Site

Supersedes GEI-100694 for Darwin Only

Instruction Guide

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Overview and Preparation

Refer to the flow chart, Non-CMS Environment Upgrade Process.

This document describes how to upgrade to ControlST software version 03.06.10C from version 03.06.07C. The upgrade process requires that the following procedures be performed in the order listed below.

- Prepare computer for installation
- Install the ControlST software upgrade files.



Attention

Install the ControlST software on all workstations, HMIs, and computers before proceeding with the next step. Do not install GE CMS Server on any other computer except the Master HMI.

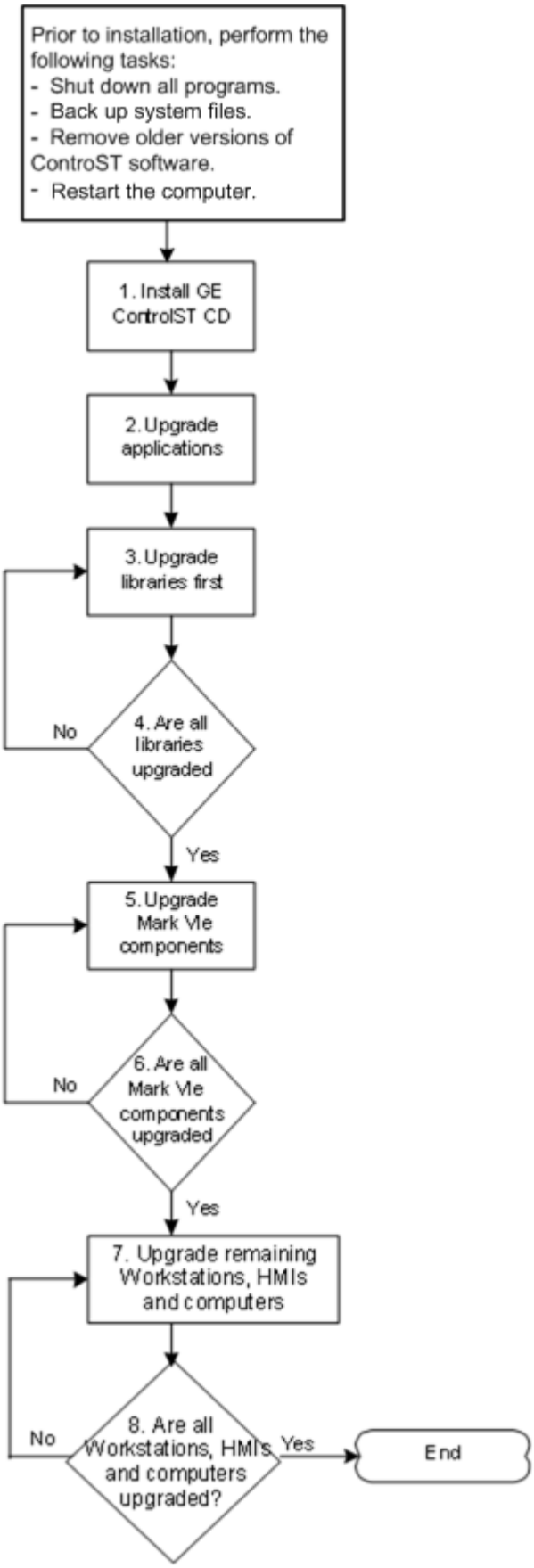
- Upgrade the block libraries.
- Upgrade the configured Mark* VIe control components.
- Upgrade the remaining workstations, HMIs, and other computers.

Note These procedures assume that the user is upgrading their existing ControlST software and has in their possession the following: new ControlST software version 03.06.10C for installation, USB Protection Device (UPD), access level privileges, and required system passwords.

Note Before beginning the upgrade process, close any programs currently open, create backup copies of the master data files, and verify that you have a minimum of 2000 MB of disk space for each version of the ControlST software installed.

➢ To prepare computer for installation

1. Shut down all programs.
2. Back up the system files.
3. Remove older versions of ControlST software including WorkstationST* and ToolboxST* applications.
4. Restart the computer.
5. Install the ControlST software. Select only the WorkstationST and ToolboxST installation packages (first two options).



Non-CMS Environment Upgrade Process

Install ControlST Software Upgrade Files

➢ **To install the upgrade files**

1. Place the ControlST disk in the CD/DVD drive. The installation starts automatically.

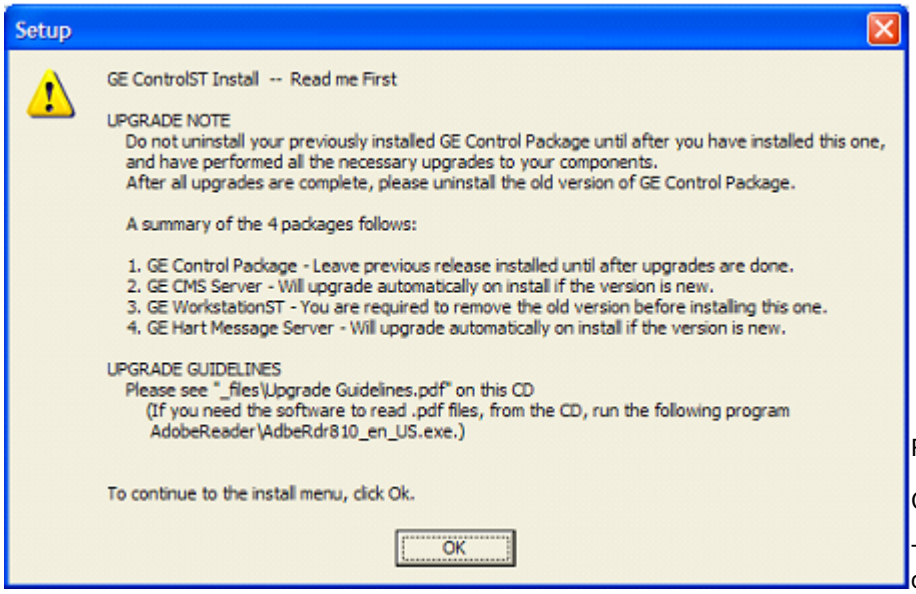
Note If the installation does not start automatically, navigate to the CD-ROM drive and run *Setup.exe*.



Attention

The ControlST installation associates .tcw files on a single machine with the ToolboxST Version Selector, which helps to determine the correct version that should be used to open the .tcw file. However, if the most recently installed ControlST version is prior to V03.03.xx, all .tcw files will be associated with that specific ControlST version.

2. The **Read Me First** message box displays.

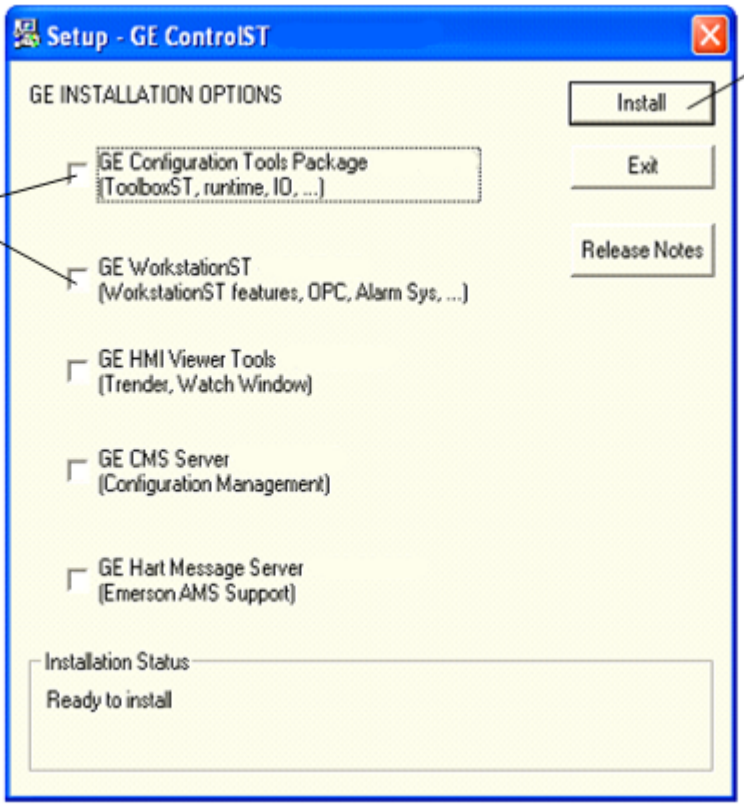


Read the **Read me First** message.

Click **OK.**

The **Setup - GE ControlST** dialog box displays.

3. From the **Setup - GE ControlST** dialog box, select the **GE Installation Options** to install.



For the Darwin site, choose the first two options only.

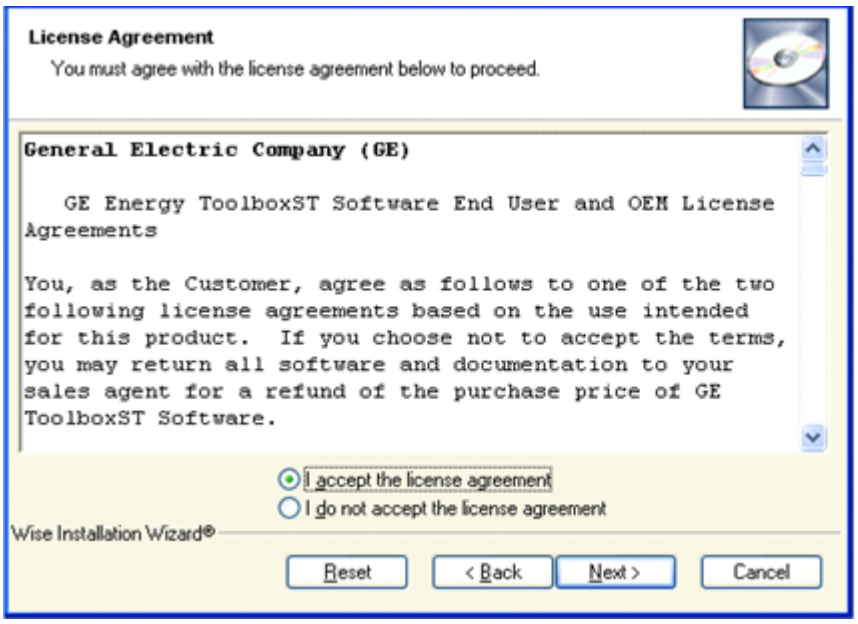
Select the required **GE INSTALLATION OPTIONS** and click **Install**.

The **Welcome** dialog box displays.



Click **Next**.

The **License Agreement** dialog box displays.



Click **I accept the license agreement**.

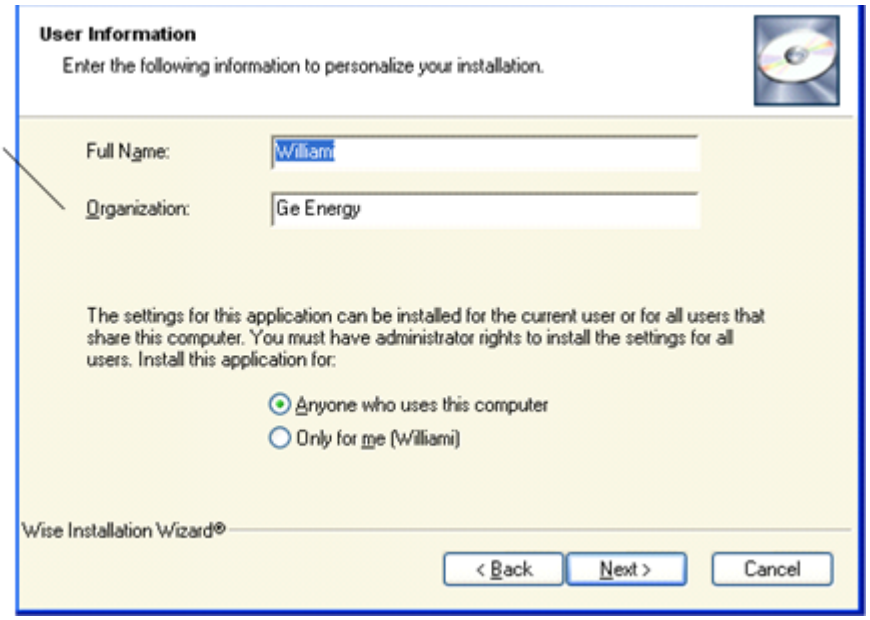
Click **Next**.

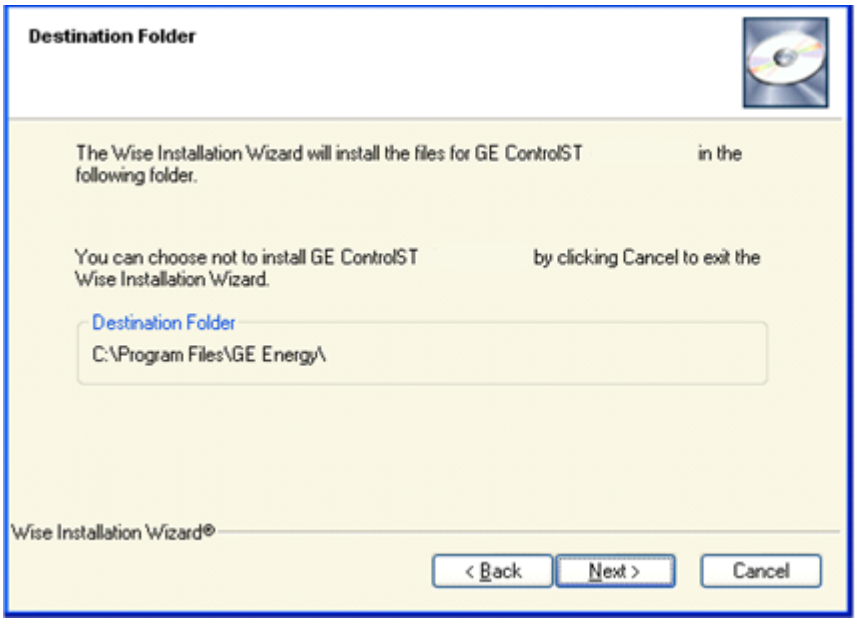
The **User Information** dialog box displays.

Enter the **Full Name** and **Organization**.

Click **Next**.

The **Destination Folder** dialog box displays.





Click **Next**.

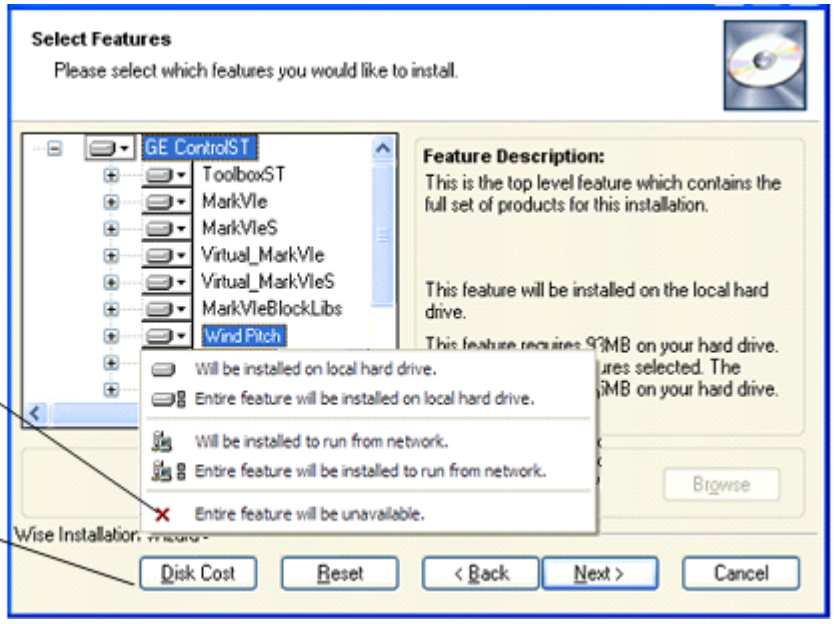
The **Select Features** dialog box displays.

All components are installed by default.

Right-click on any feature NOT to be installed and select **Entire feature will be unavailable**.

Click the **Disk Cost** button to display the amount of disk space necessary for a feature.

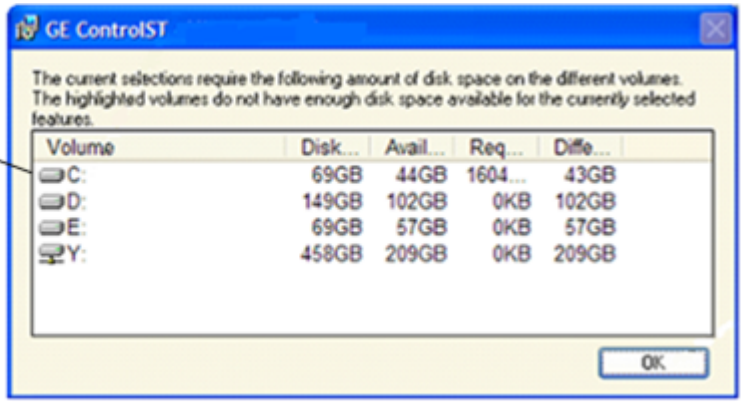
Click **Next**.

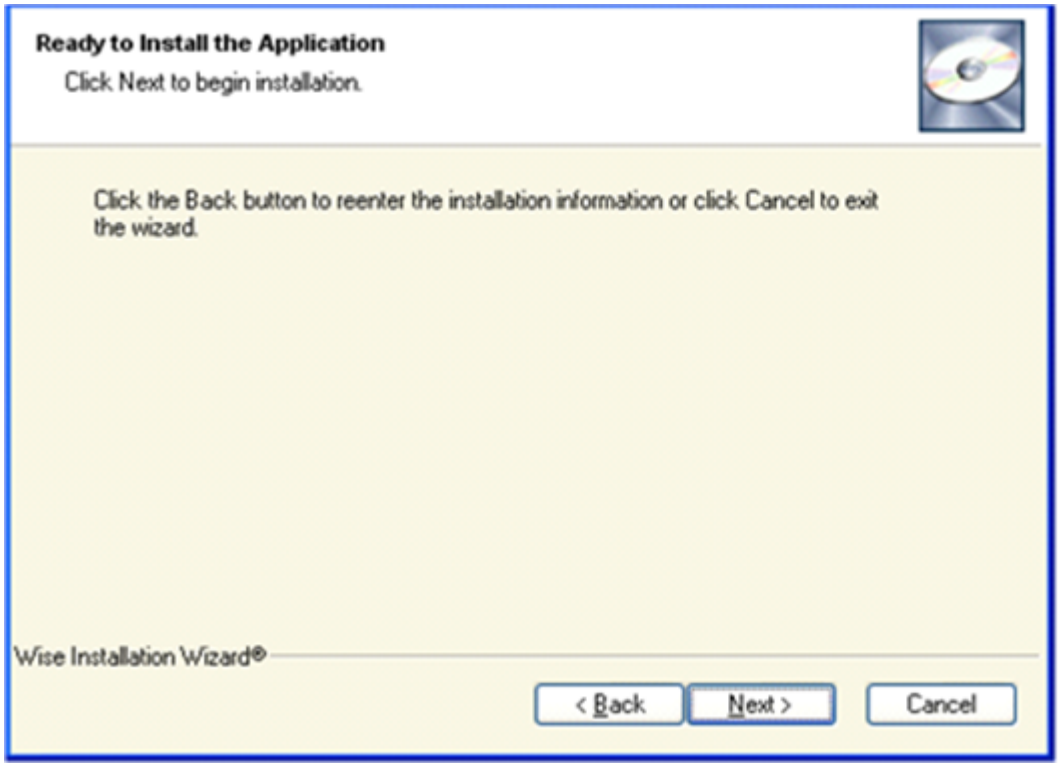


The software is typically installed on the **C:** drive.

Click **OK**.

The **Ready to Install the Application** dialog box displays.





Click **Next**.

A progress indicator displays the status.

4. When installation of the GE Configuration Tools Package completes, click **Finish.**

Note PROFIBUS is automatically installed at this time. This will only occur when the ToolboxST application is being installed.

5. The installation process repeats for the GE WorkstationST package. Follow the wizard screens to complete the installation.
6. After the installation completes, exit the setup.
7. Restart the computer.

Upgrade System Components

The ControlST software suite supports a variety of control equipment. These products include updated firmware for the equipment and configuration tools that are available as component editors inside the ToolboxST application. To use the new versions, each component must be upgraded. Follow the procedures in this section to upgrade system components in this order:

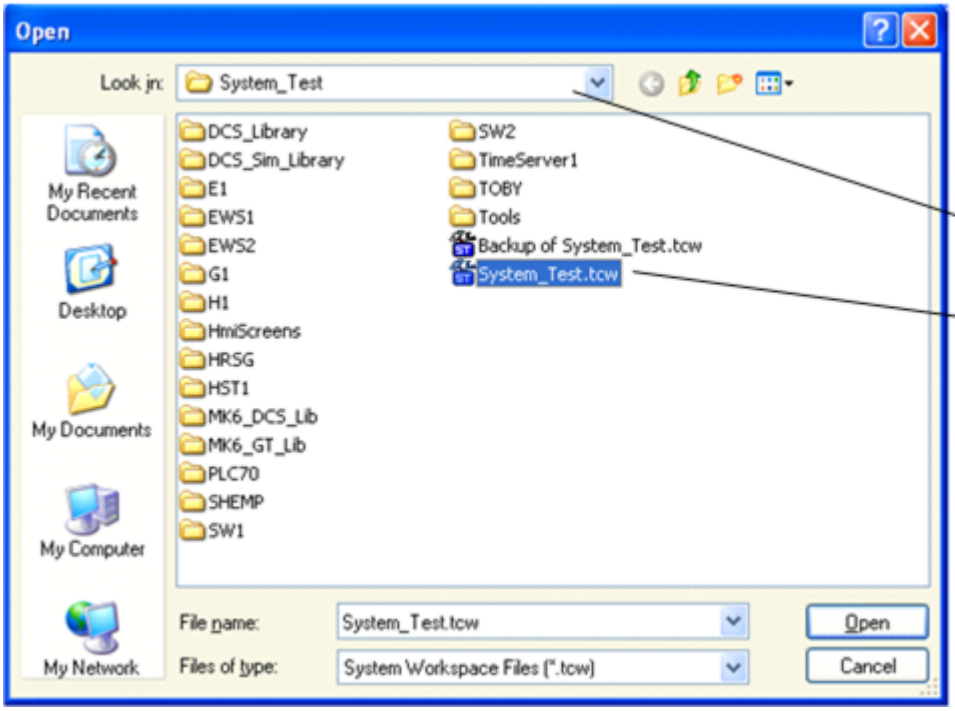
- Block libraries
- Mark VIe components
- Workstations, HMIs, and other computers

Block Libraries

➢ **To upgrade block libraries**

1. From the **Start** menu, select **All Programs**, **GE ControlST**, **ToolboxST**, **ToolboxST** to start the **ToolboxST** application.

Note When the ToolboxST application starts, it opens to the last system.tcw file that was used. If this is not the system that you want to upgrade, navigate to the location of the system.tcw file to upgrade.



From the **System Editor File** menu, select **Open System**.

Locate and select the system (**System_Test**) file.

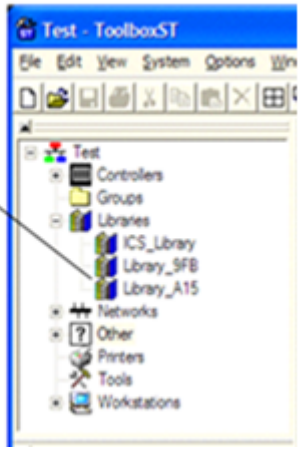
Select the **.tcw** (**System_Test.tcw**) file and click **Open**.



Attention

Libraries must be upgraded first.

From the **Tree View**, double-click a **Library** item.

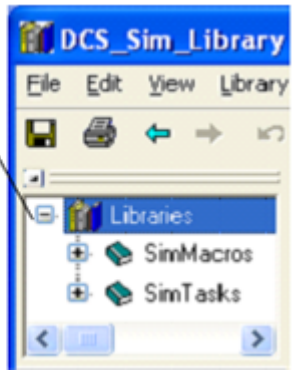


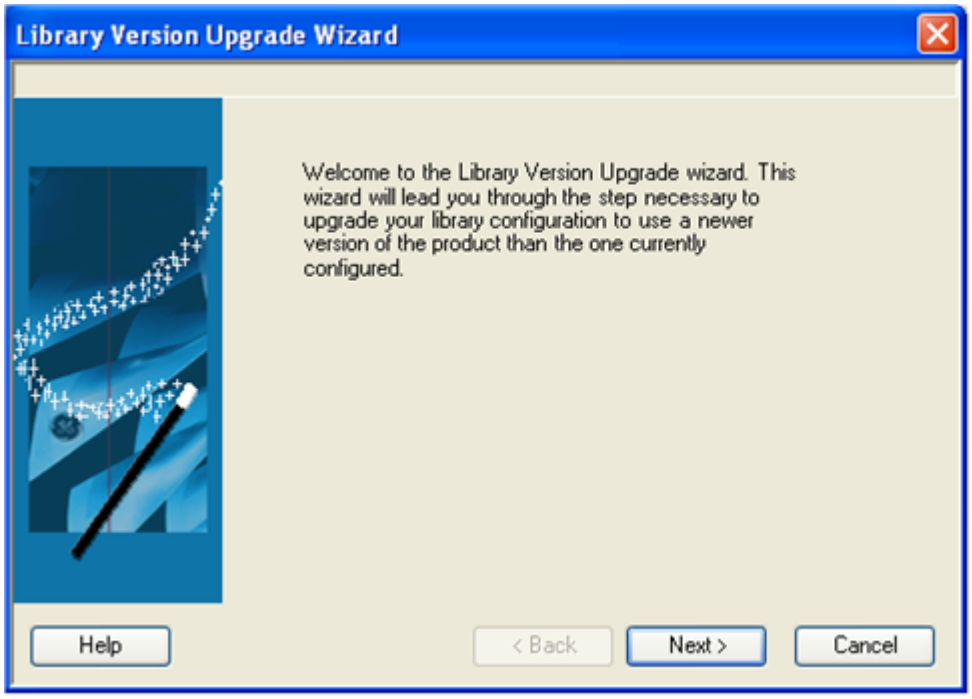
2. If a ToolboxST message about version displays, click **OK**.

From the **Component Editor Tree View**, highlight the **Libraries** item.

From the **File** menu, select **Upgrade**.

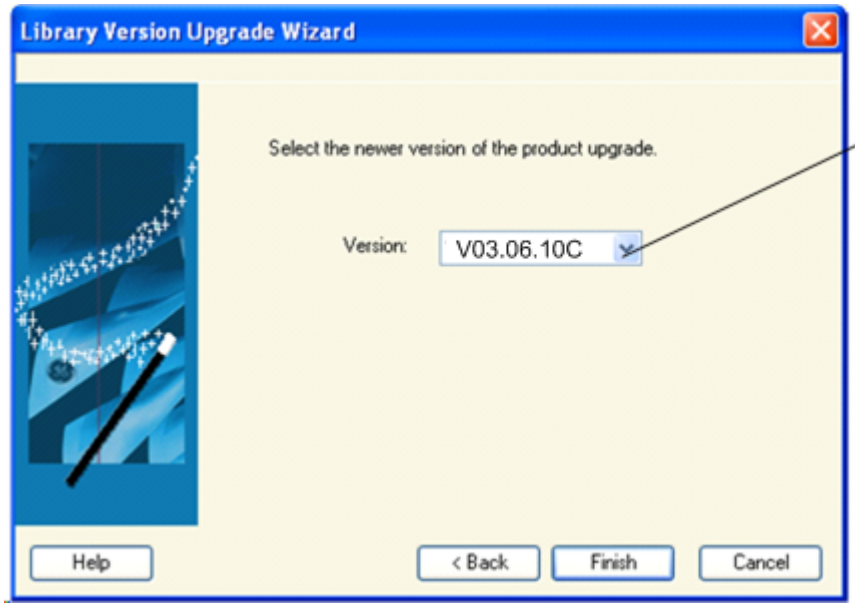
The **Library Version Upgrade Wizard** displays.





Click **Next**.

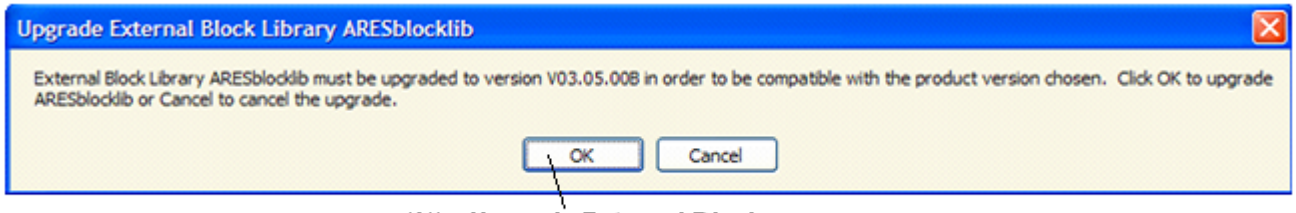
The following dialog box displays.



Verify the latest **Version** is selected from the drop-down list.

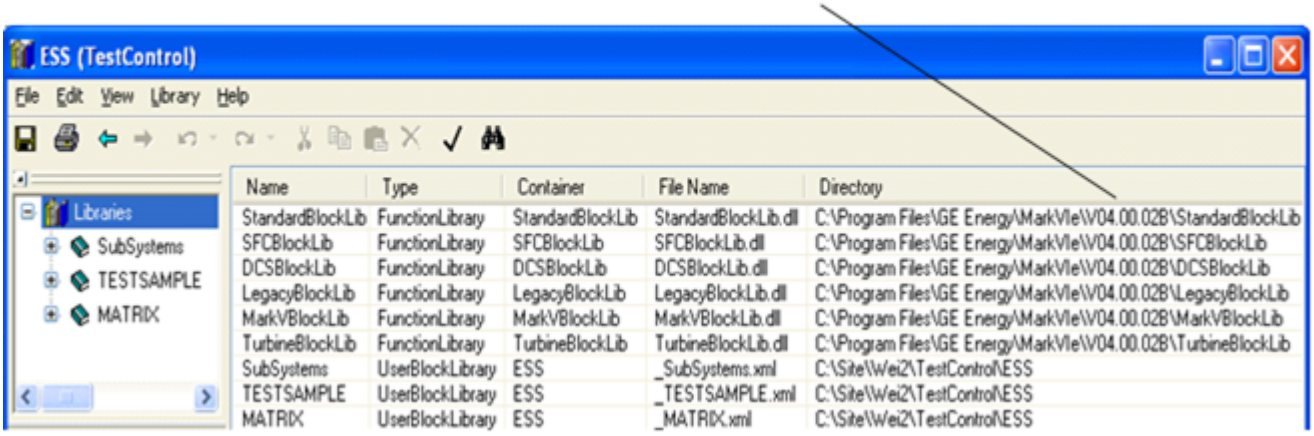
Click **Finish**.

Some libraries may contain references to optional external libraries, such as ARESBlockLib and MPCBlockLib. The Product Version Wizard upgrades them automatically when prompted.

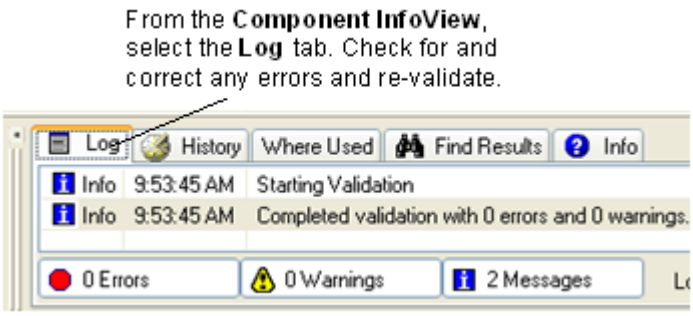


If the **Upgrade External Block Library** dialog box displays, click **OK**.

From the **Tree View**, select Libraries. Verify the **BlockLibs** are upgraded by reviewing their version references.



3. From the toolbar, click **Save** to save the library upgrades.
4. A ToolboxST message about version displays. Click **OK**.
5. With the Library container still open, from the toolbar, click **Validate** to validate the changes.



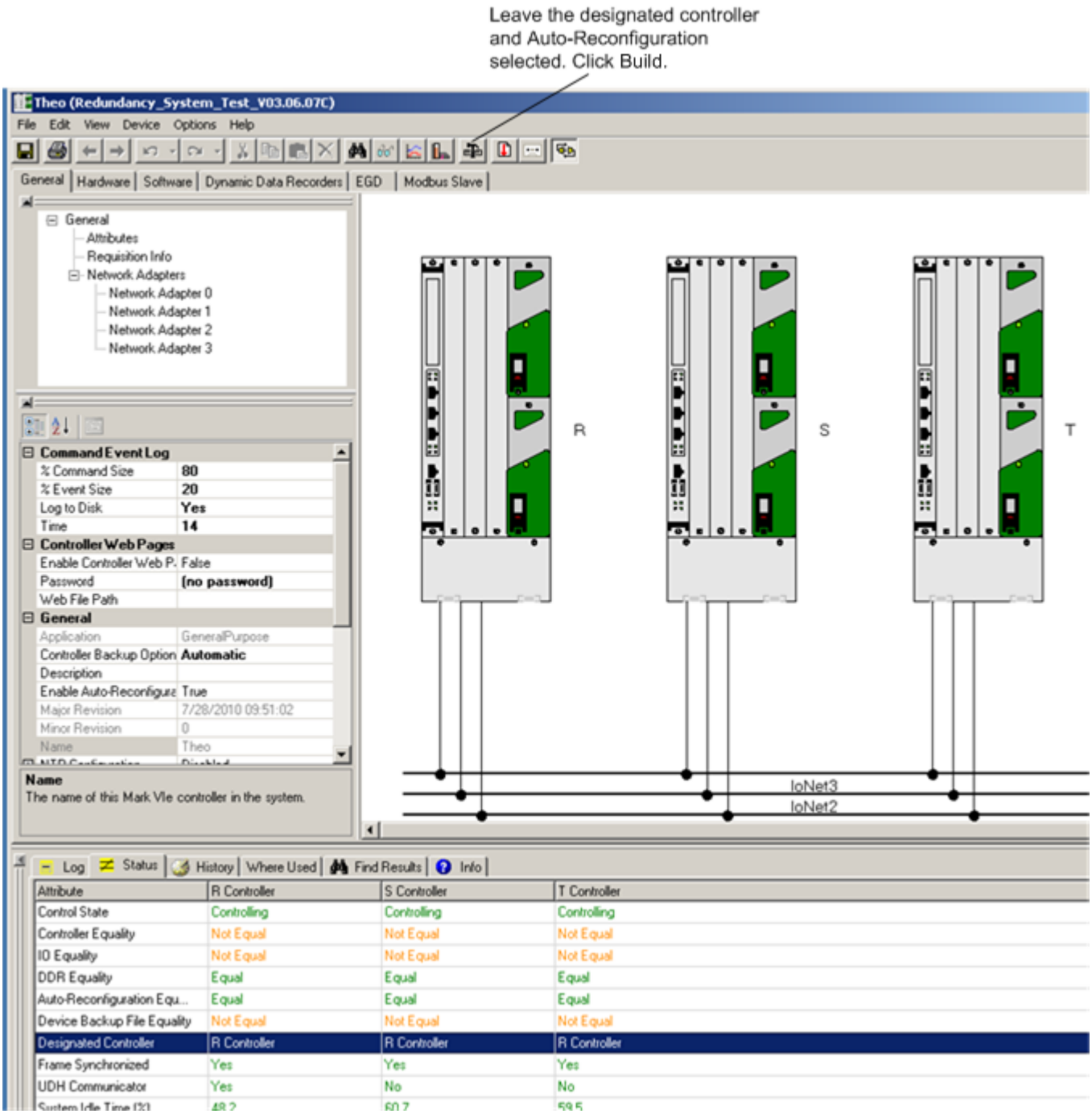
6. From the **File** menu, select **Close** to close the library.
7. Repeat this procedure for the remaining library containers displayed in the **Tree View**.

➢ **To upgrade the Mark VIe controller and I/O modules**

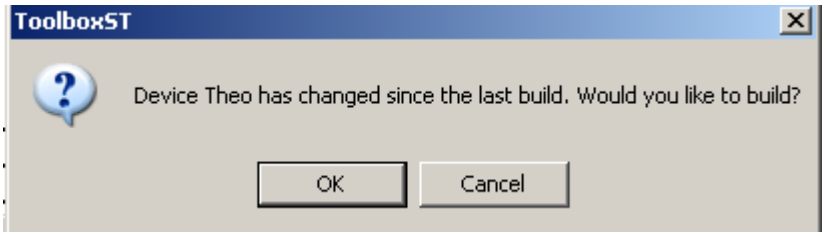
Attention

Before performing an online download to the controllers, verify that there are no diagnostic or process alarms present. If there are, these should be resolved before performing the software upgrade. These could result in a turbine trip when downloading I/O pack firmware.

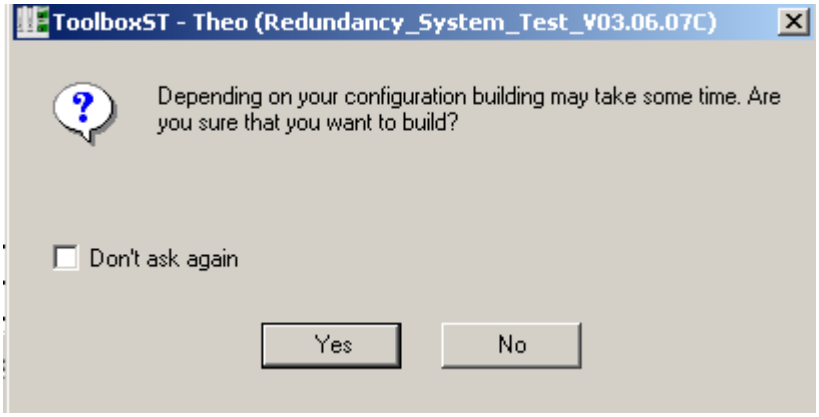
1. From the **System Editor** double-click the Mark VIe controller. The **Component Editor** displays.



2. Follow the message boxes and wizard pages to complete the build.



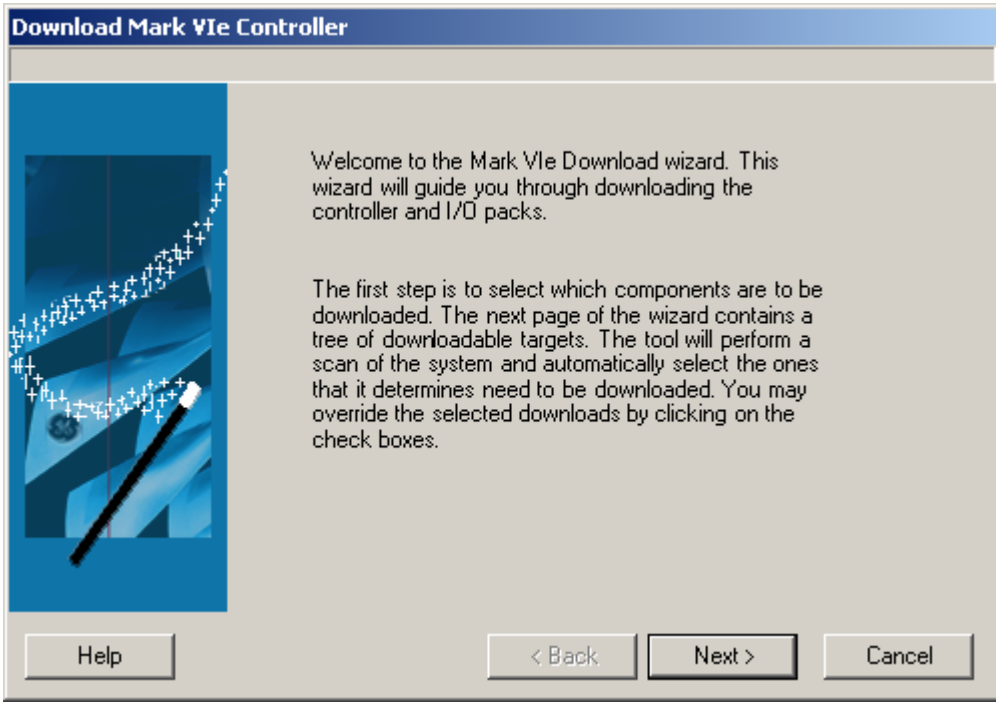
Click **OK.**



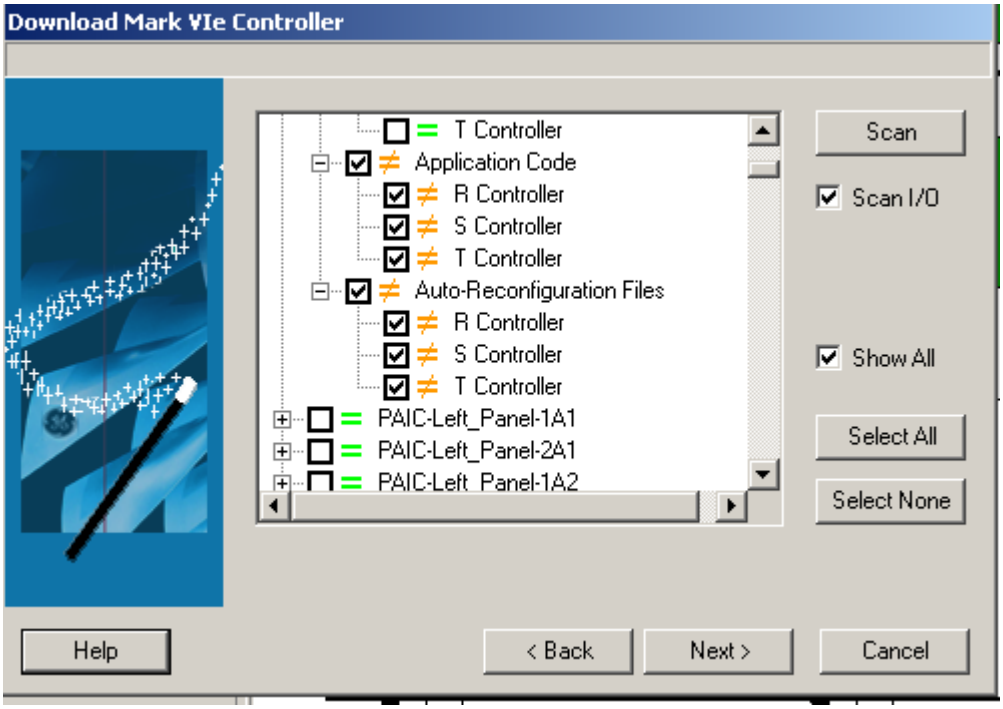
Click **Yes.**



Click **OK.**



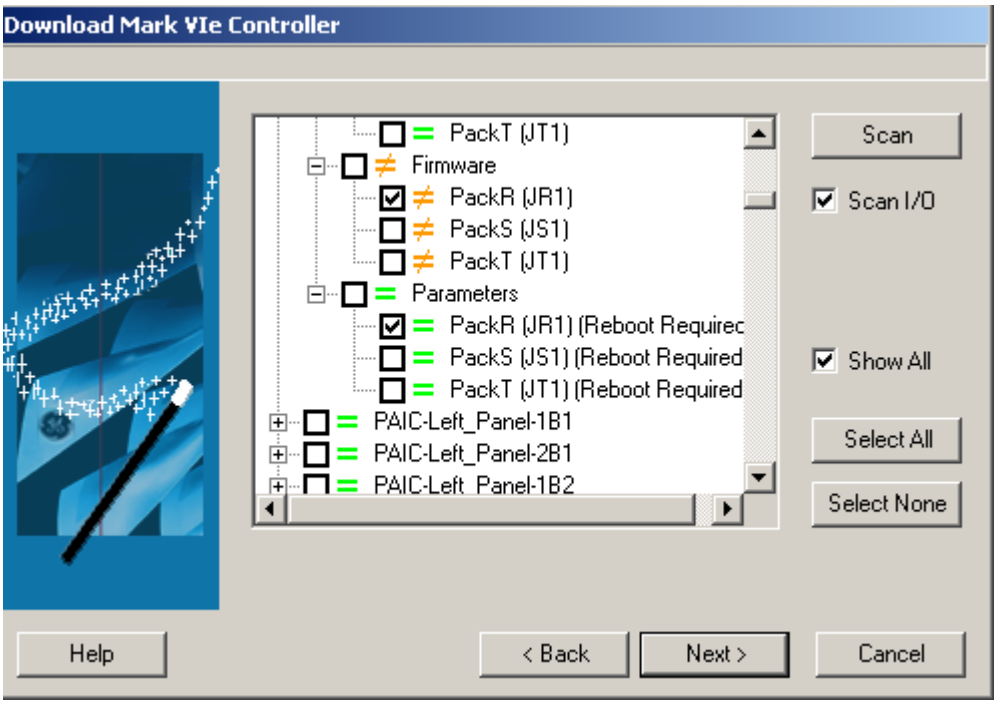
Click **Next.**





Attention

Select the **Application Code** and **Auto-Reconfiguration Files.**

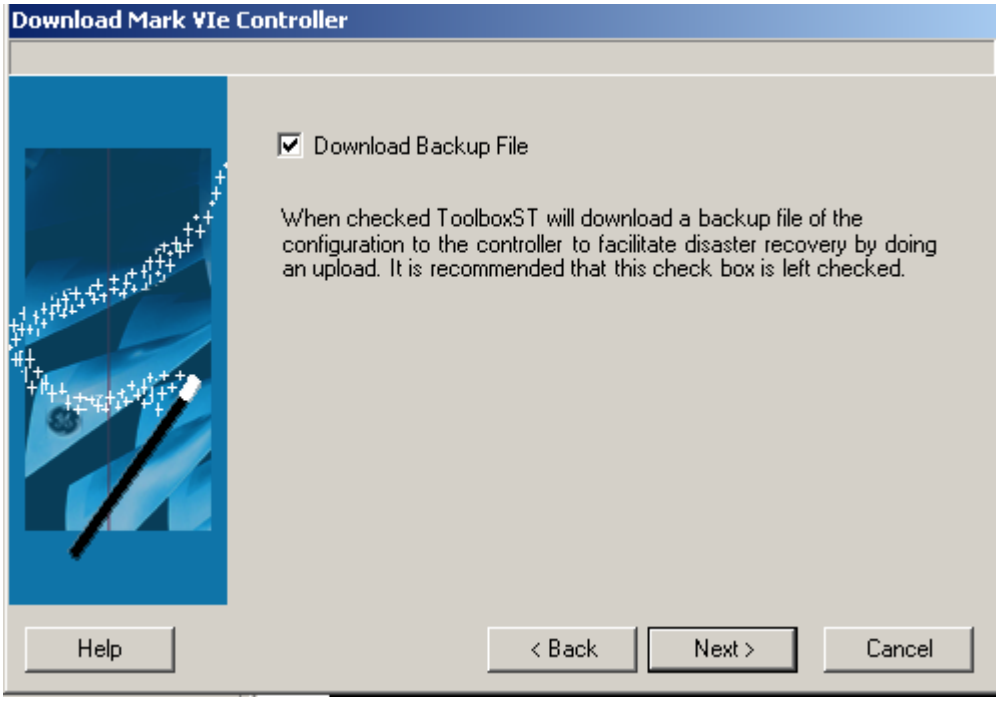




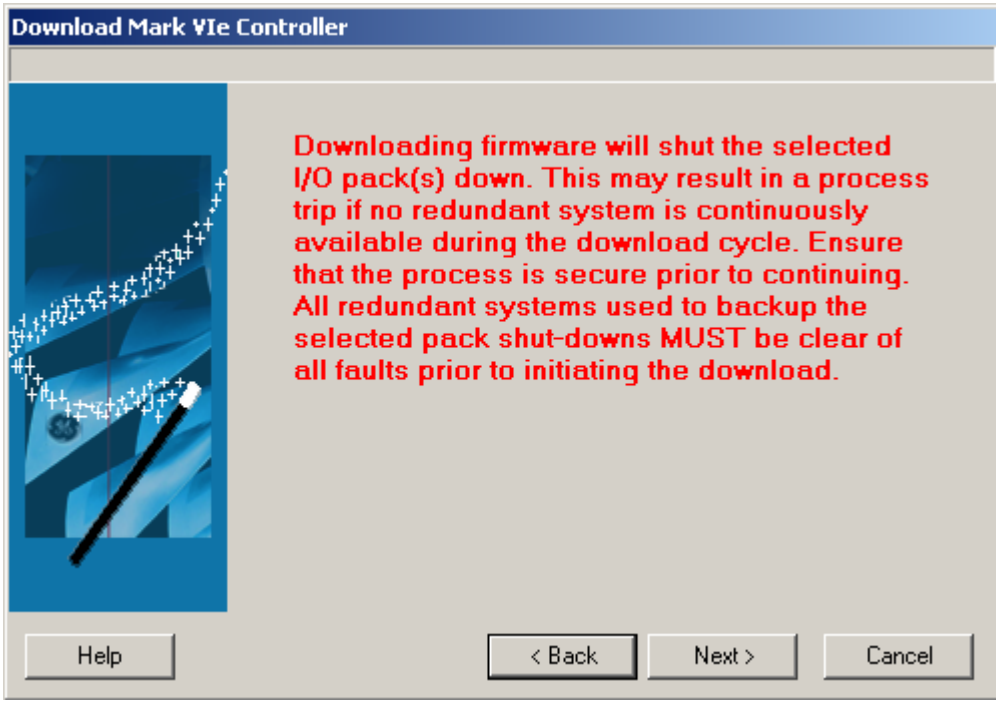
Attention

Scroll down and select only the **PackR** Firmware and Parameters for each PDIA I/O pack.

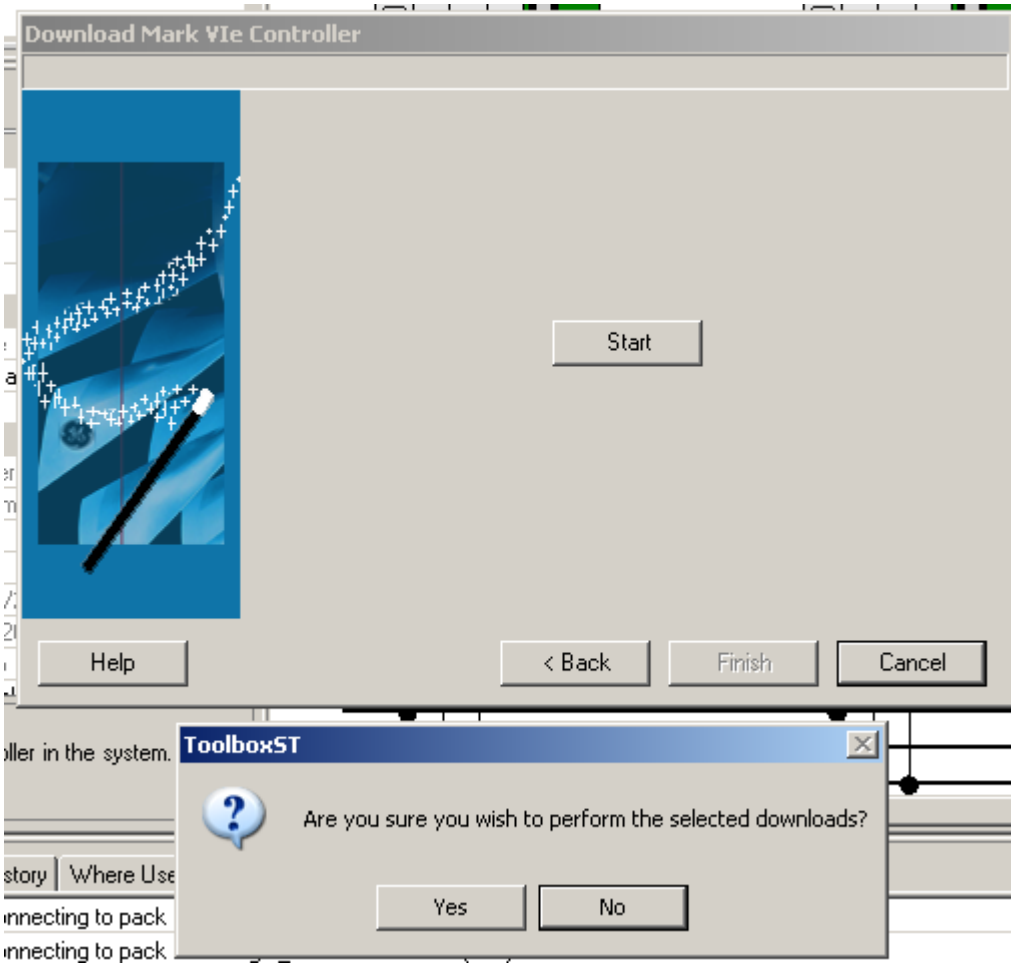
Click **Next.**



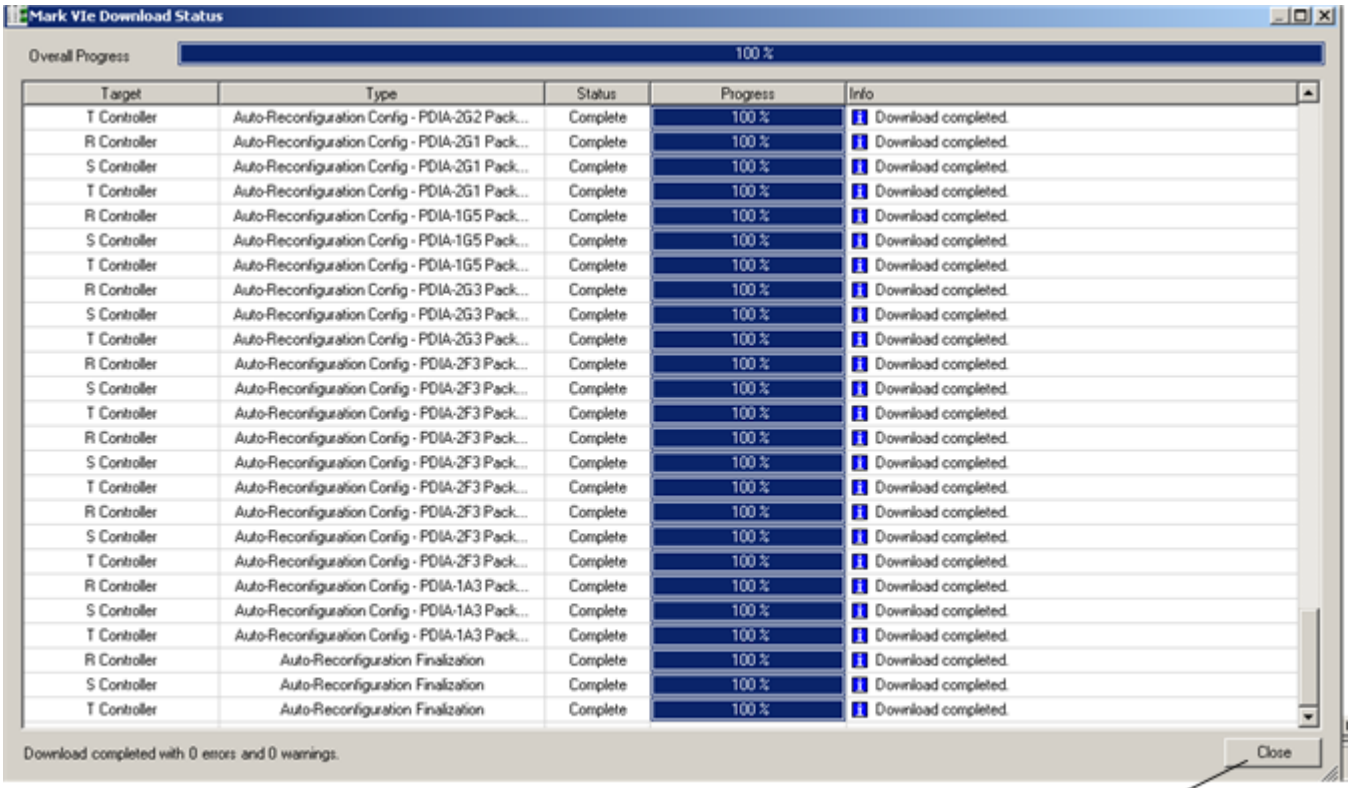
Click **Next.**



Click **Next.**

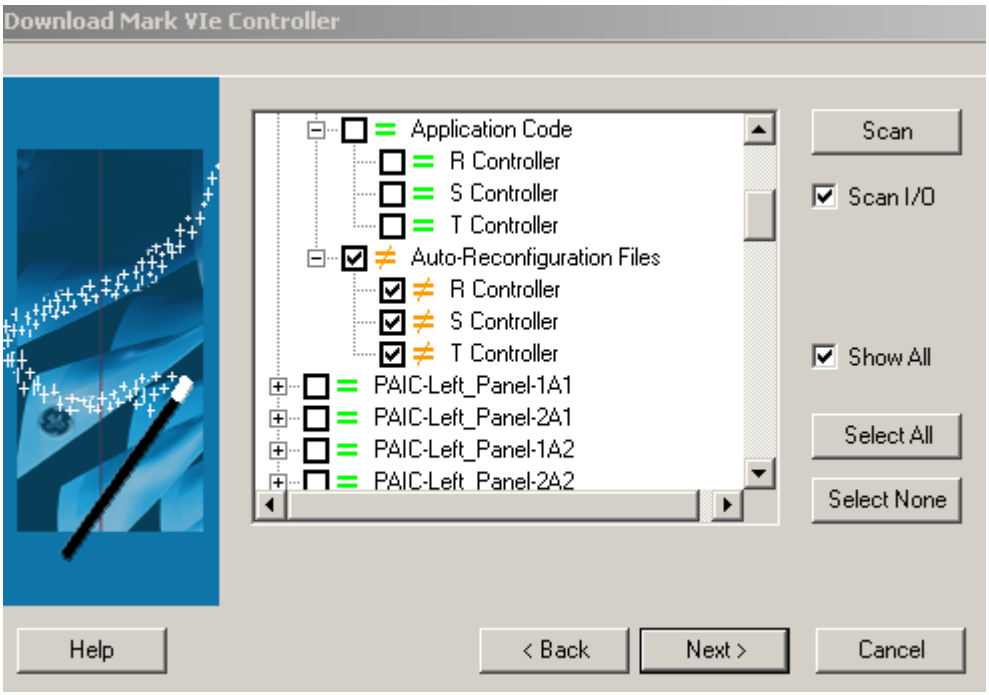


Click **Yes.**



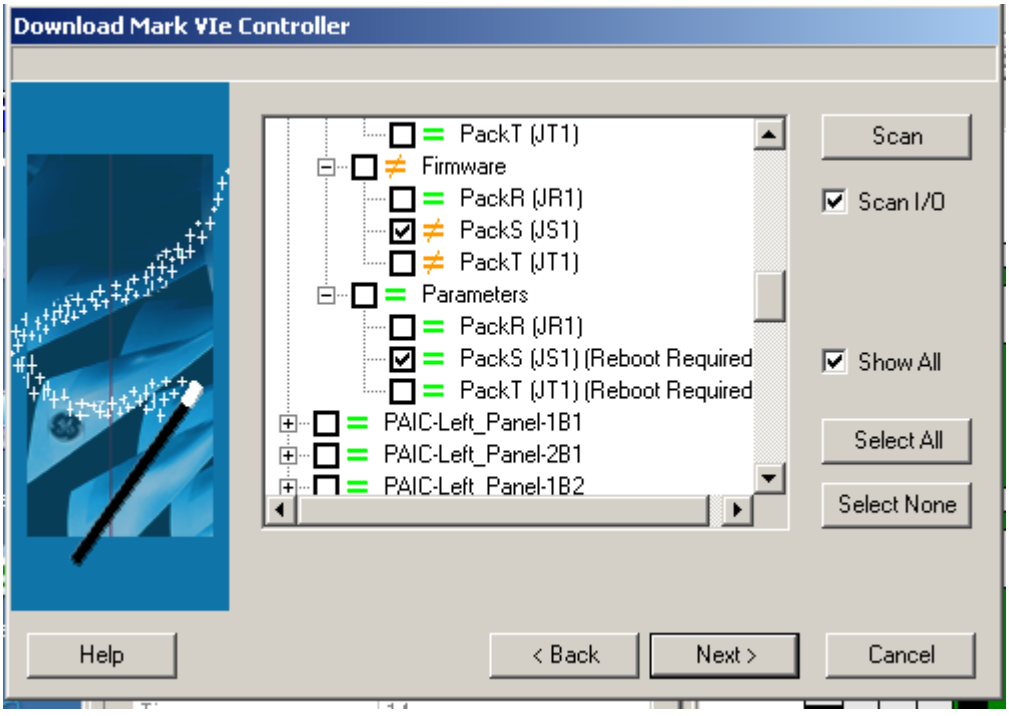
After the download completes, click **Close**.

3. Wait five minutes for all changes to take affect. Allow the I/O packs to come back to their controlling state.
4. From the toolbor, click the **Download** button.
5. Follow the wizard screens except this time, verify that **Auto-Reconfiguration Files** remain selected, then scroll down and select only **PackS** Firmware and Parameters for each PDIA I/O pack.



Attention

Ensure **Auto-Reconfiguration Files** are selected.



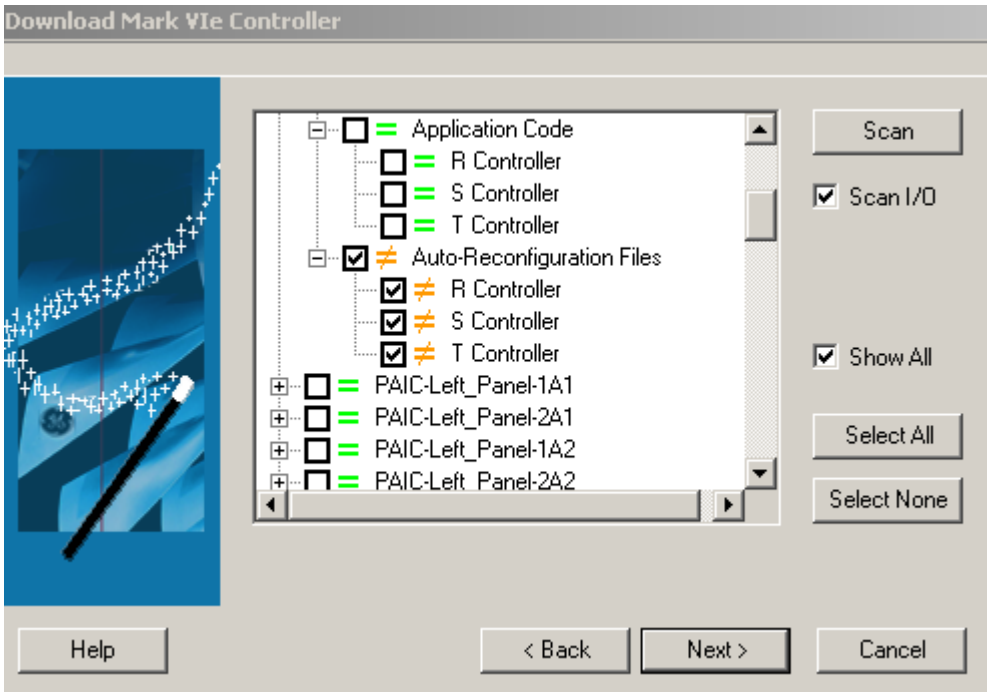
Attention

Scroll down and select only **PackS** Firmware and Parameters for each PDIA I/O pack.

Click **Next.**

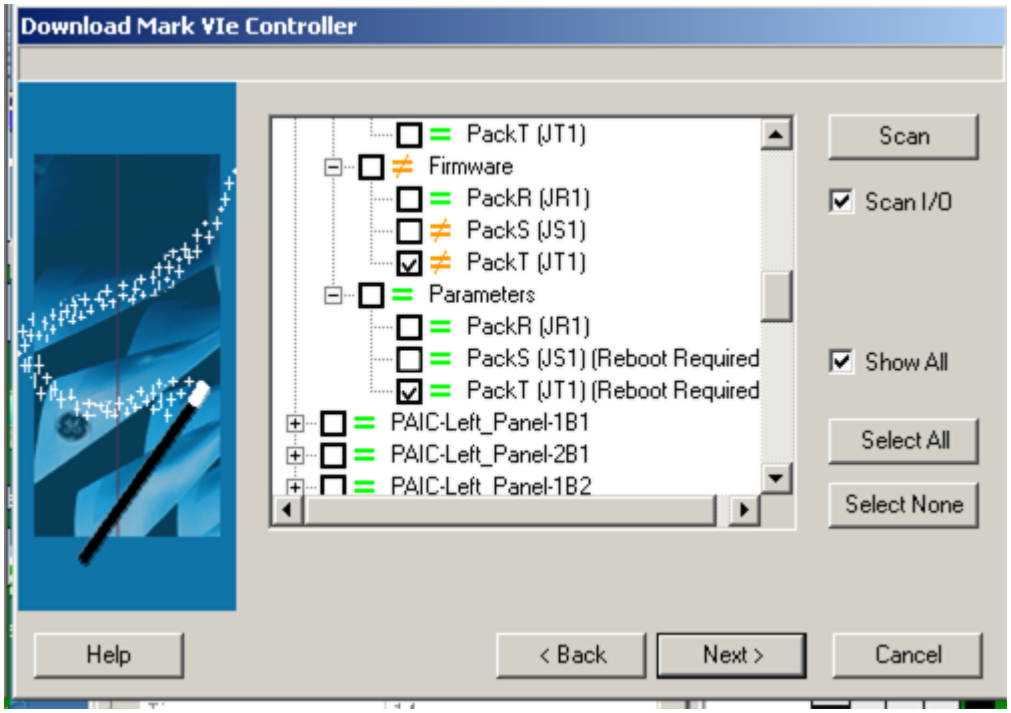
6. Wait five minutes for all changes to take affect. Allow the I/O packs to come back to their controlling state.
7. From the toolbar, click the **Download** button.

8. Follow the wizard screens except this time, verify that **Auto-Reconfiguration Files** remain selected, then scroll down and select only **PackT** Firmware and Parameters for each PDIA I/O pack.



Attention

Ensure **Auto-Reconfiguration Files** are selected.



Attention

Scroll down and select only **PackT** Firmware and Parameters for each PDIA I/O pack.

Click **Next.**

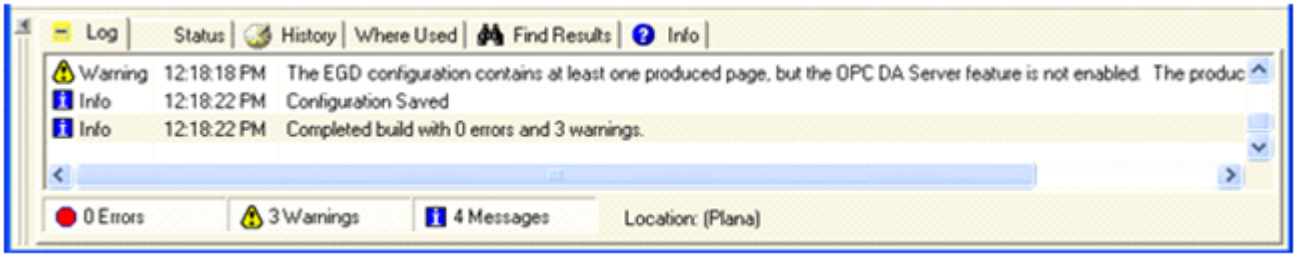
9. Wait five minutes for all changes to take affect. Allow the I/O packs to come back to their controlling state.

Workstations, HMIs, and other Computers

After all libraries and components have been upgraded, use the following procedure to upgrade the workstations and other computers in the system.

➢ To upgrade workstations, HMIs, and other computers

1. To upgrade other computers, refer to the section Install ControlST Software Upgrade Files.
2. Repeat step 1 for each workstation, HMI, or computer.
3. To upgrade a configured computer, open the updated ***.tcw** file from the Master Computer.
4. From the **System Editor Tree View**, double-click a computer.
5. A ToolboxST message about the older version displays, click **OK.** The **Component Editor** displays.
6. From the toolbar, click **Build**  .
7. From the **Component Editor**, select the **Log** tab. Correct any errors and click **Build** again.
8. A ToolboxST version message displays. Click **OK.**
9. From the toolbar, click **Download**  to download the configuration.
10. Use the **Workstation Download wizard** to download the configuration.
11. From the **Log** tab, verify that there are no errors and check any warning messages.



12. Click **Connect**  to verify the configuration is equal.
13. Save and **Close** the workstation component.
14. Repeat steps 5 through 13 for each workstation, HMI, or computer.

Subsequent updates on other workstations, HMIs, and computers can be accomplished by checking out the Latest Version of the system.tcw file from the Master HMI.

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